

Quick Start Guide

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Basic Concepts

1. Linux - The "Operating System" of MutoRS
2. Docker - Can be simply understood as a lightweight virtual machine
3. ROS2 - A communication framework, the "nervous system" of MutoRS

This tutorial will briefly introduce the three most important basic concepts in the MutoRS learning process (Linux, Docker, ROS2). If you want to get started with MutoRS quickly, this section will help you get a fast, simple, and preliminary understanding of these three points.

Basic Concepts

1. Linux - The "Operating System" of MutoRS

In simple terms: Just as Windows or macOS is the operating system for a computer, Linux is the operating system installed on the MutoRS's main controller, the Raspberry Pi.

The system running on the Raspberry Pi is actually Raspberry Pi OS (formerly Raspbian), the official operating system for the Raspberry Pi, which is based on the Debian Linux distribution.

Q: What is the relationship between Linux, Debian, Ubuntu, and Raspberry Pi OS? Are they all operating systems? Which one actually runs on the Raspberry Pi?

- A: Yes, all of these can be called operating systems. Their relationship is hierarchical, not parallel (though Ubuntu and Raspberry Pi OS can be considered parallel).

Imagine the process of building a house:

Linux is like the foundation and main structure of the house - it's the core part that makes the whole system run, but you can't live in it with just that.

Debian is like a simple and practical house built on this foundation - very stable, has everything, but the decoration is relatively simple.

Ubuntu is like a beautifully decorated modern apartment - it's based on the Debian house but has been redecorated, adding many modern facilities, making it look prettier and more convenient to use.

Raspberry Pi OS is like a custom-decorated mini-house designed specifically for the "small apartment" that is the Raspberry Pi - also based on Debian, but optimized for small spaces, removing unnecessary things and adding many special facilities suitable for a small home.

Basics we need to know before starting with MutoRS:

- Basic universal file operations on Linux (e.g., `cd` to change directory)
- Simple command-line operations (the `sudo` command, as editing some startup scripts requires administrator privileges)

- Remotely connecting to MutoRS (SSH, please read the [04, Remote Login Operation] section for details)

2. Docker - Can be simply understood as a lightweight virtual machine

Why do we need it:

- For easy and fast portability. For example, if we have a MutoRS project (including system environment configuration, required prerequisite software, MutoRS example code, etc.), we need it to run on a Raspberry Pi, a Jetson board, or even a computer. Without Docker, we would have to install and configure the MutoRS running environment on each platform separately, which would be very troublesome. With Docker, we just need to package the MutoRS running environment (which uses Ubuntu 22.04 + ROS2 Humble) into a Docker image and then load and use it on other machines.

Knowledge needed to get started with MutoRS

- How to start and stop a container
 - Enter `start_muto_docker` in the terminal to start the Docker container
 - In another terminal, enter `enter_muto_docker` to get inside the container to operate
- Basic container management (can be skipped for a quick start)

3. ROS2 - A communication framework, the "nervous system" of MutoRS

In simple terms: ROS2 is like the nervous system of MutoRS, allowing its various components (like LiDAR, depth camera, etc.) to communicate and coordinate with each other.

Basics you need to know:

- How to start a ROS2 program (usually with the `ros2 launch` command, as you will see in subsequent tutorials)
- Some common ROS2 debugging commands (`ros2 topic list`, `ros2 node list`, `ros2 service list`, `ros2 param list`, etc., used to check the status of the currently running ROS2 program)